

Project Master Profile: Malaysian Neural Voice API

1. Executive Summary

A self-hosted Text-to-Speech API delivering natural-sounding, cloned, and fine-tuned voices in Malaysian languages (Bahasa Malaysia, English, Mandarin, Tamil, and regional accents). Designed as the first locally-owned high-quality voice infrastructure in Malaysia, targeting businesses that require custom, controllable, and low-latency voice synthesis — without foreign API dependency.

2. Core Technology Stack

- **TTS Engine:** XTTS v2 (`coqui-ai/TTS`) — open-source, cross-lingual voice cloning & fine-tuning.
- **Model Size:** ~1.6B parameters; supports zero-shot cloning from a 6-second sample.
- **Languages (out-of-the-box):** 17+ including English, Malay, Mandarin, Tamil, Arabic, Japanese, Korean, French.
- **Fine-tuning:** QLoRA / full fine-tune on custom Malay datasets for accent accuracy and domain terminology.
- **API Framework:** FastAPI (Python 3.11+), with auto-generated Swagger documentation.
- **Containerisation:** Docker + optional NVIDIA Container Toolkit for portability.
- **Audio Processing:** librosa, pydub (format conversion, silence trimming).
- **Authentication:** API-key based, managed via environment variables.

3. Recommended Hardware Specifications

A. Cloud Instance (for prototyping / low-volume API)

- **GPU:** NVIDIA A10, A4000, or equivalent — minimum **16 GB VRAM** (24 GB preferred for batch inference).
- **vCPU:** 8 cores.
- **RAM:** 32 GB.
- **Storage:** 100 GB SSD (model weights ~2 GB; audio output storage per request negligible).
- **Estimated cost:** RM 2.50 – RM 6.00 per hour on Vast.ai / RunPod / Paperspace.

B. Self-Owned Server (for production, high volume, zero marginal cost)

A single build from a RM10,000 budget. Prices as of mid-2026 Malaysia.

| Component | Recommendation | Reason |

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| **GPU** | NVIDIA RTX 3090 24 GB (used) or RTX 4070 Ti Super 16 GB (new) | 24 GB VRAM handles large batches & fine-tuning comfortably. 3090 best price/performance. |

| **CPU** | AMD Ryzen 7 7700X or Intel i7-13700K | Fast single-core for API latency; multi-core for audio post-processing. |

| **Motherboard** | AM5 (B650) / LGA1700 (B760) — PCIe 4.0 support | Stable power delivery for GPU. |

| **RAM** | 32 GB DDR5 (2×16 GB) 5200 MHz+ | Ample for loading model + concurrent requests. |

| **Storage** | 1 TB NVMe SSD (Gen4, e.g., Samsung 980 Pro / WD Black SN850X) | Fast read for model weights; stores fine-tuned checkpoints and audio assets. |

| **PSU** | 850W Gold-rated (e.g., Corsair RM850e / Seasonic Focus) | 3090 transient spikes require headroom. |

| **Cooling & Case** | Mesh-front airflow case + aftermarket CPU cooler | Prevents thermal throttling during long training runs. |

| **Total Build Cost** | ~RM 8,000 – RM 9,800 | Leaves balance for UPS or backup drive. |

Running cost: Electricity ~RM 150–250/month (based on 500W average draw, 24/7). Zero per-inference charge.

4. Voice Model & Languages

- **Baseline Model:** XTTS v2 (pre-trained, English-centric).
- **Phase 1 Voice:** Boss's own voice, fine-tuned on 30–60 minutes of clean Malay speech.
- **Phase 2 Voice Library:** 5–10 curated voices (male, female, senior, child) covering Bahasa Malaysia, Manglish, Mandarin, and Tamil.
- **Custom Voices:** Enterprise clients can provide 1+ hour of clean data for a dedicated fine-tuned model deployed as a private endpoint.

5. API Architecture

- **Endpoint:** `POST /v1/tts`
- **Input JSON:** `{ "text": "string", "speaker\_id": "string", "language": "string", "speed": float }`
- **Output:** `audio/wav` or `audio/mpeg`, 24 kHz sample rate.
- **Additional endpoints (future):** `/v1/speakers` (list available voices), `/v1/clone` (temporary voice cloning upload), `/v1/fine-tune` (enterprise fine-tune job submission).

- **Latency target:** <2 seconds for a 30-word sentence (real-time factor <1 on GPU).
- **Concurrency:** Handled via FastAPI async + GPU batching (when implemented).

6. Development Phases

Phase 1 — Foundation (Weeks 1–4)

- Deploy XTTS v2 on designated hardware.
- Fine-tune first voice with Boss's dataset.
- Build & document `/v1/tts` endpoint.
- Internal testing of latency and pronunciation accuracy.

Phase 2 — Voice Expansion (Weeks 5–8)

- Produce and fine-tune 5+ standard voices.
- Implement speaker gallery endpoint.
- Basic analytics and rate limiting.
- Start limited alpha with friendly businesses.

Phase 3 — Enterprise & Scale (Months 3–6)

- Build self-service clone & fine-tune portal.
- Optimisation: batching, streaming TTS, serverless auto-scaling (multi-GPU).
- Launch commercial tiers (monthly subscriptions, pay-per-use, enterprise on-premise deployment).

7. Budget Allocation (RM10,000 Cap)

- **Hardware (own server):** RM 7,850 – RM 9,850
- **Reserve (electricity, incidentals):** RM 150 – RM 2,150
- **Cloud prototyping (optional test before purchase):** RM 300 – RM 500 (one-month trial)

8. Scalability & Migration Path

- Identical software stack runs on cloud or bare-metal.
- Moving from cloud to self-owned server: transfer Docker images, model checkpoints, and environment config – no code changes.
- Scaling beyond one GPU: load balancing with multiple nodes possible using same API

container.

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